0580_w18_qp_42_Q11

11 **(a)** $\mathbf{a} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$ $\mathbf{b} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$ $\mathbf{c} = \begin{pmatrix} 14 \\ 9 \end{pmatrix}$

(i) Find $3\mathbf{a} - 2\mathbf{b}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ [2]

(ii) Find $|\mathbf{a}|$.

..... [2]

(iii) $m\mathbf{a} + n\mathbf{b} = \mathbf{c}$

Write down two simultaneous equations and solve them to find the value of m and the value of n. Show all your working.

$m =$ .....

$n =$ ..... [5]

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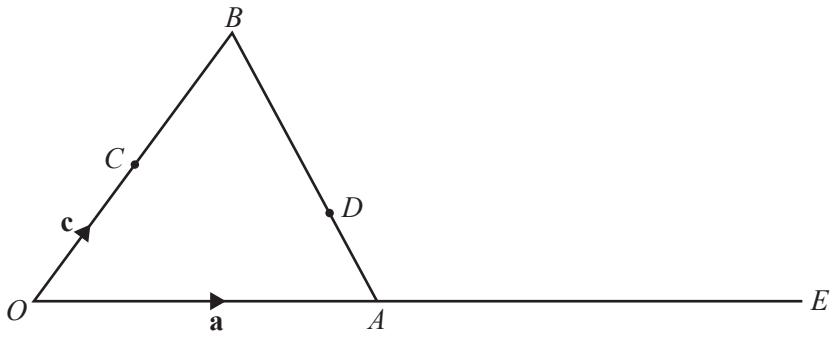
(b)



NOT TO SCALE

OAB is a triangle and C is the mid-point of OB.
D is on AB such that $AD : DB = 3 : 5$.
OAE is a straight line such that $OA : AE = 2 : 3$.
$\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$.

(i) Find, in terms of **a** and **c**, in its simplest form,

(a) $\overrightarrow{AB}$,

$\overrightarrow{AB} =$ ..... [1]

(b) $\overrightarrow{AD}$,

$\overrightarrow{AD} =$ ..... [1]

(c) $\overrightarrow{CE}$,

$\overrightarrow{CE} =$ ..... [1]

(d) $\overrightarrow{CD}$.

$\overrightarrow{CD} =$ ..... [2]

(ii) $\overrightarrow{CE} = k\overrightarrow{CD}$

Find the value of k.

$k =$ ..... [1]